# TidyTuesday Week 30

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Thursday 23rd July, 2026

## Contents

## 1 Weekly Summary

**AI-Generated Content**

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to the only weekly column brave enough to shove two economic time series into a blender and act shocked when a smoothie comes out. As always, our beloved dartboard selected three pairs of variables that have absolutely no business being in the same spreadsheet, and as always, we ran the numbers anyway. Standard disclaimer, delivered in my most soothing voice: these are randomly paired series, nobody is claiming anyone causes anyone, and if you build a portfolio around this newsletter you deserve what happens next.

Monday gave us the Fed's holdings of medium-term Treasury securities versus a discontinued consumer price index for services in the South, Size Class D – a series so obscure and so retired it hasn't posted since 2017. The raw regression came out gasping "significant!" with an R-squared of 0.77, and I was briefly tempted to write a think piece. Then we detrended it, the R-squared slumped to 0.37, and the slope politely flipped sign from positive to negative, which is the statistical equivalent of a witness changing their entire story on the stand. Translation: most of that "relationship" was just two things drifting through time together like strangers on the same escalator. A trend mirage, gently evaporating on contact. Deeply moving. Not real.

Tuesday, however, is where I have to reluctantly stop being sarcastic for approximately one sentence. We paired "Withdrawals from Income of Quasi-Corporations" (a phrase that sounds like a threat) against depreciation of nonresidential structures, and the raw fit was a frankly obnoxious R-squared of 0.98. My eyes rolled so hard I strained something – surely another shared-trend fraud. But no: after detrending, it still posted an R-squared of 0.87 with a p-value that requires scientific notation to even write down, and the slope barely budged from 0.42 to 0.40. That is the rare, genuinely interesting case where the relationship survives having its trend surgically removed. Is there a real link? Plausibly – both are dollar-denominated measures of nonfinancial business activity, so they may actually be tracking the same underlying economic thing rather than just aging in parallel. I refuse to be excited, but I am, quietly, in a corner.

And then Wednesday arrived to restore balance to the universe by regressing the population of Nuckolls County, Nebraska against a medical price index for congenital anomalies, because sure, why not. The raw fit strutted in at 0.88 with a confident negative slope, suggesting that as one Nebraska county empties out, the cost of certain medical services climbs – a correlation so gloriously unrelated it belongs in a museum. Detrended, the R-squared collapsed to 0.36, though I'll note the slope stubbornly stayed negative and significant-ish, so this one is at least a more dignified mirage than Monday's. Still: two things going opposite directions over decades is not a relationship, it's a coincidence with good posture. See you next week, when the dartboard once again humbles us all.

## 2 Date: 2026-07-21

**Series ID: TREAS1T5**

This series is titled Assets: Securities Held Outright: U.S. Treasury Securities: Maturing in over 1 Year to 5 Years: Wednesday Level and has a frequency of Weekly, As of Wednesday. The units are Millions of U.S. Dollars and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 2002-12-18 and the observation end date is 2026-07-15.The popularity of this series is 21.

**Series ID: CUURD300SAS**

This series is titled Consumer Price Index for All Urban Consumers: Services in South - Size Class D (DISCONTINUED) and has a frequency of Monthly. The units are Index 1982-1984=100 and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1977-12-01 and the observation end date is 2017-12-01.The popularity of this series is 1.

### 2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

| | | | |
|---|---|---|---|
| **Dep. Variable:** | value_fred_CUURD300SAS | **R-squared:** | 0.774 |
| **Model:** | OLS | **Adj. R-squared:** | 0.765 |
| **Method:** | Least Squares | **F-statistic:** | 82.38 |
| **Date:** | Tue, 21 Jul 2026 | **Prob (F-statistic):** | 3.15e-09 |
| **Time:** | 13:11:10 | **Log-Likelihood:** | -104.86 |
| **No. Observations:** | 26 | **AIC:** | 213.7 |
| **Df Residuals:** | 24 | **BIC:** | 216.2 |
| **Df Model:** | 1 | | |
| **Covariance Type:** | nonrobust | | |

| | **coef** | **std err** | **t** | **P> \|t\|** | **[0.025** | **0.975]** |
|---|---|---|---|---|---|---|
| **const** | 214.5017 | 4.768 | 44.985 | 0.000 | 204.660 | 224.343 |
| **value_fred_TREAS1T5** | 6.405e-05 | 7.06e-06 | 9.076 | 0.000 | 4.95e-05 | 7.86e-05 |

| | | | |
|---|---|---|---|
| **Omnibus:** | 1.003 | **Durbin-Watson:** | 0.294 |
| **Prob(Omnibus):** | 0.606 | **Jarque-Bera (JB):** | 0.795 |
| **Skew:** | -0.004 | **Prob(JB):** | 0.672 |
| **Kurtosis:** | 2.143 | **Cond. No.** | 1.16e+06 |

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

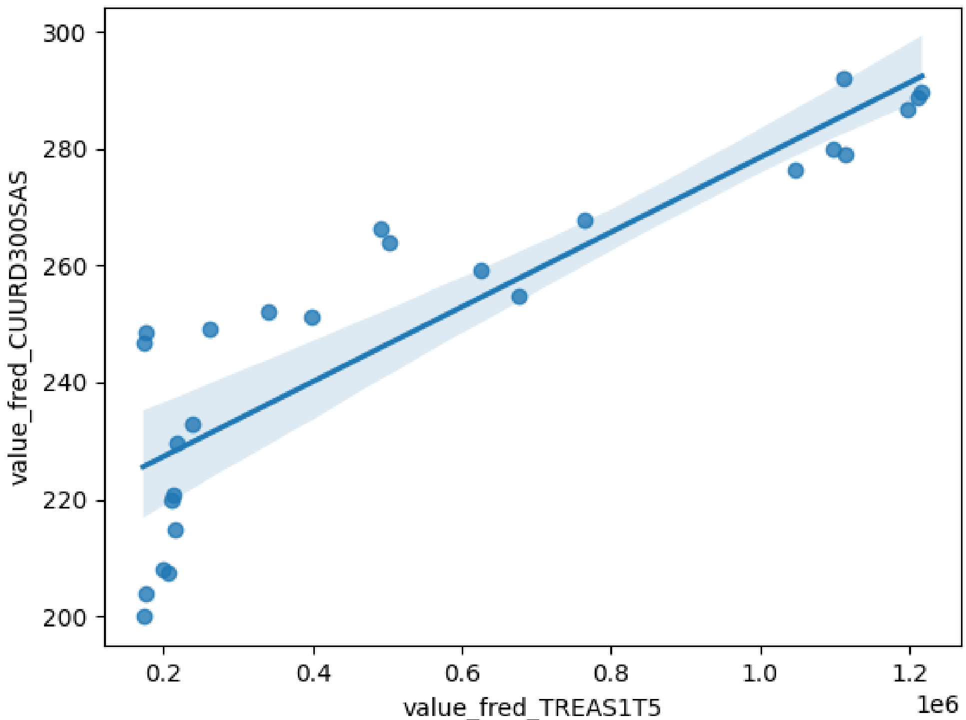


**Figure 1:** Raw Regression Plot for 2026-07-21

[2] The condition number is large, 1.16e+06. This might indicate that there are strong multicollinearity or other numerical problems.

### 2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

| | | | |
|---|---|---|---|
| **Dep. Variable:** | value_fred_CUURD300SAS_detrended | **R-squared:** | 0.365 |
| **Model:** | OLS | **Adj. R-squared:** | 0.339 |
| **Method:** | Least Squares | **F-statistic:** | 13.82 |
| **Date:** | Tue, 21 Jul 2026 | **Prob (F-statistic):** | 0.00107 |
| **Time:** | 13:11:11 | **Log-Likelihood:** | -63.094 |
| **No. Observations:** | 26 | **AIC:** | 130.2 |
| **Df Residuals:** | 24 | **BIC:** | 132.7 |
| **Df Model:** | 1 | | |
| **Covariance Type:** | nonrobust | | |

| | **coef** | **std err** | **t** | **P> \|t\|** | **[0.025** | **0.975]** |
|---|---|---|---|---|---|---|
| **const** | 4.318e-12 | 0.559 | 7.72e-12 | 1.000 | -1.154 | 1.154 |
| **value_fred_TREAS1T5_detrended** | -1.309e-05 | 3.52e-06 | -3.718 | 0.001 | -2.04e-05 | -5.83e-06 |

| | | | |
|---|---|---|---|
| **Omnibus:** | 0.194 | **Durbin-Watson:** | 1.006 |
| **Prob(Omnibus):** | 0.907 | **Jarque-Bera (JB):** | 0.403 |
| **Skew:** | 0.028 | **Prob(JB):** | 0.818 |
| **Kurtosis:** | 2.393 | **Cond. No.** | 1.59e+05 |

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.59e+05. This might indicate that there are
strong multicollinearity or other numerical problems.

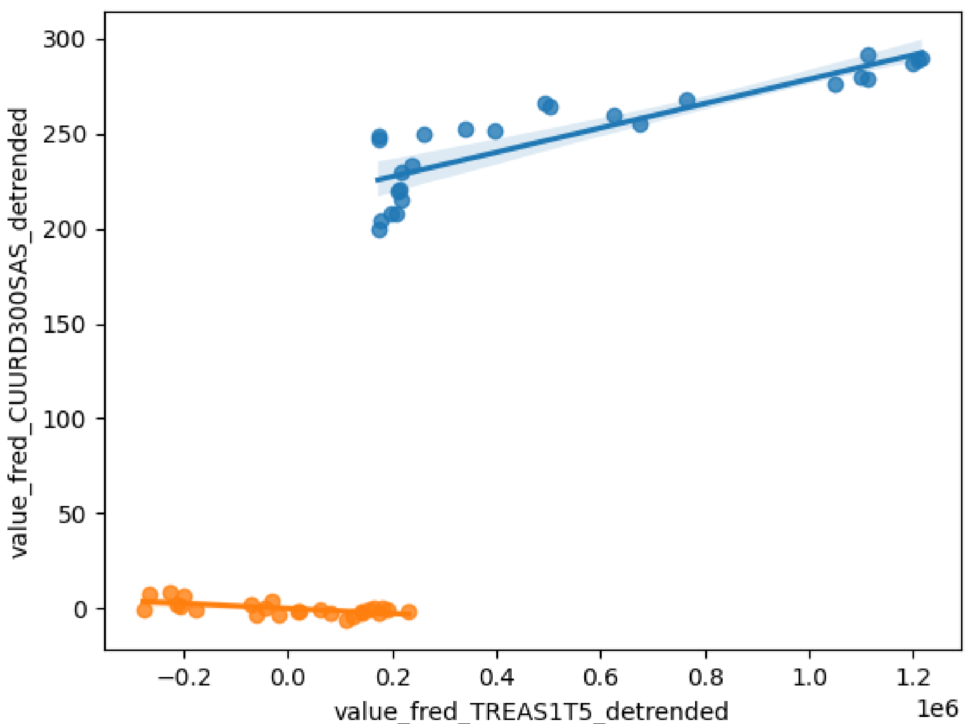


**Figure 2:** Detrended Regression Plot for 2026-07-21

# 3 Date: 2026-07-22

**Series ID: NNBICPA027N**

This series is titled Nonfinancial Noncorporate Business; Withdrawals from Income of Quasi-Corporations, Paid, Transactions and has a frequency of Annual, End of Period. The units are Millions of U.S. Dollars and the seasonal adjustment is Seasonally Adjusted Annual Rate.The observation start date is 1959-01-01 and the observation end date is 2024-01-01.The popularity of this series is 1.

**Series ID: M1YTOTL1ST000**

This series is titled Current-Cost Depreciation of Private and government fixed assets: Nonresidential: Structures and has a frequency of Annual. The units are Millions of Dollars and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1925-01-01 and the observation end date is 2024-01-01.The popularity of this series is 1.

## 3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

| | | | |
|---|---|---|---|
| **Dep. Variable:** | value_fred_M1YTOTL1ST000 | **R-squared:** | 0.985 |
| **Model:** | OLS | **Adj. R-squared:** | 0.984 |
| **Method:** | Least Squares | **F-statistic:** | 4111. |
| **Date:** | Wed, 22 Jul 2026 | **Prob (F-statistic):** | 8.67e-60 |
| **Time:** | 13:16:17 | **Log-Likelihood:** | -777.64 |
| **No. Observations:** | 66 | **AIC:** | 1559. |
| **Df Residuals:** | 64 | **BIC:** | 1564. |
| **Df Model:** | 1 | | |
| **Covariance Type:** | nonrobust | | |

| | **coef** | **std err** | **t** | **P> \|t\|** | **[0.025** | **0.975]** |
|---|---|---|---|---|---|---|
| **const** | -1720.1069 | 5808.925 | -0.296 | 0.768 | -1.33e+04 | 9884.554 |
| **value_fred_NNBICPA027N** | 0.4156 | 0.006 | 64.115 | 0.000 | 0.403 | 0.429 |

| | | | |
|---|---|---|---|
| **Omnibus:** | 7.252 | **Durbin-Watson:** | 0.477 |
| **Prob(Omnibus):** | 0.027 | **Jarque-Bera (JB):** | 10.958 |
| **Skew:** | 0.289 | **Prob(JB):** | 0.00417 |
| **Kurtosis:** | 4.911 | **Cond. No.** | 1.31e+06 |

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

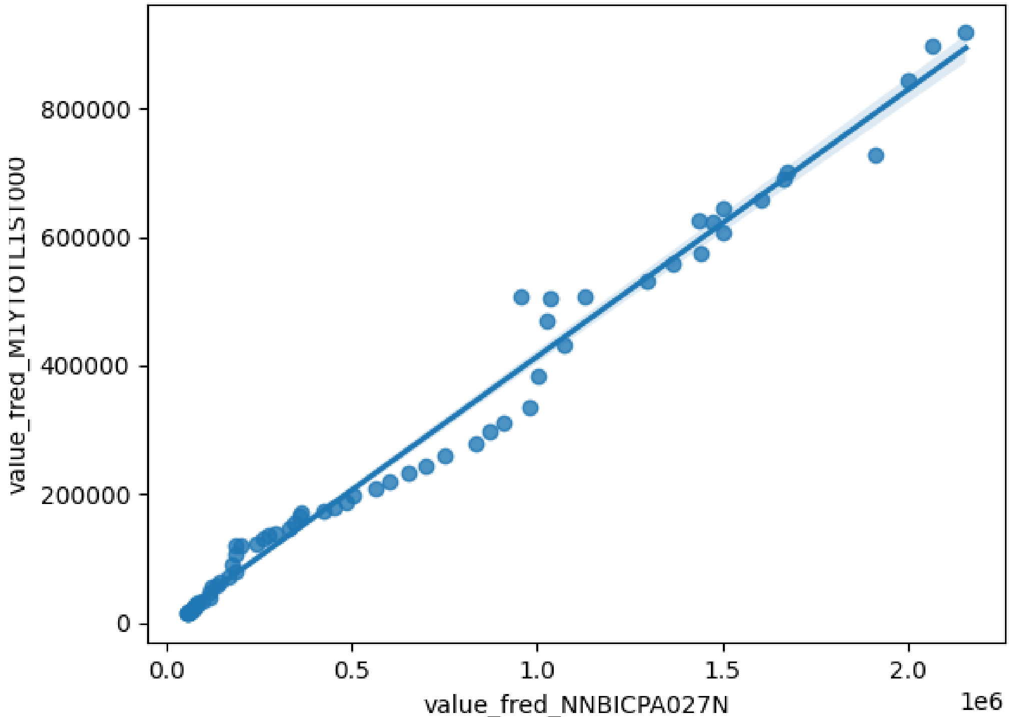


**Figure 3:** Raw Regression Plot for 2026-07-22

[2] The condition number is large, 1.31e+06. This might indicate that there are strong multicollinearity or other numerical problems.

### 3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

| | | | |
|---|---|---|---|
| **Dep. Variable:** | value_fred_M1YTOTL1ST000_detrended | **R-squared:** | 0.873 |
| **Model:** | OLS | **Adj. R-squared:** | 0.871 |
| **Method:** | Least Squares | **F-statistic:** | 439.3 |
| **Date:** | Wed, 22 Jul 2026 | **Prob (F-statistic):** | 2.31e-30 |
| **Time:** | 13:16:17 | **Log-Likelihood:** | -777.37 |
| **No. Observations:** | 66 | **AIC:** | 1559. |
| **Df Residuals:** | 64 | **BIC:** | 1563. |
| **Df Model:** | 1 | | |
| **Covariance Type:** | nonrobust | | |

| | **coef** | **std err** | **t** | **P> \|t\|** | **[0.025** | **0.975]** |
|---|---|---|---|---|---|---|
| **const** | 2.698e-09 | 3943.730 | 6.84e-13 | 1.000 | -7878.507 | 7878.507 |
| **value_fred_NNBICPA027N_detrended** | 0.4024 | 0.019 | 20.960 | 0.000 | 0.364 | 0.441 |

| | | | |
|---|---|---|---|
| **Omnibus:** | 5.762 | **Durbin-Watson:** | 0.463 |
| **Prob(Omnibus):** | 0.056 | **Jarque-Bera (JB):** | 8.362 |
| **Skew:** | 0.128 | **Prob(JB):** | 0.0153 |
| **Kurtosis:** | 4.725 | **Cond. No.** | 2.05e+05 |

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 2.05e+05. This might indicate that there are
strong multicollinearity or other numerical problems.

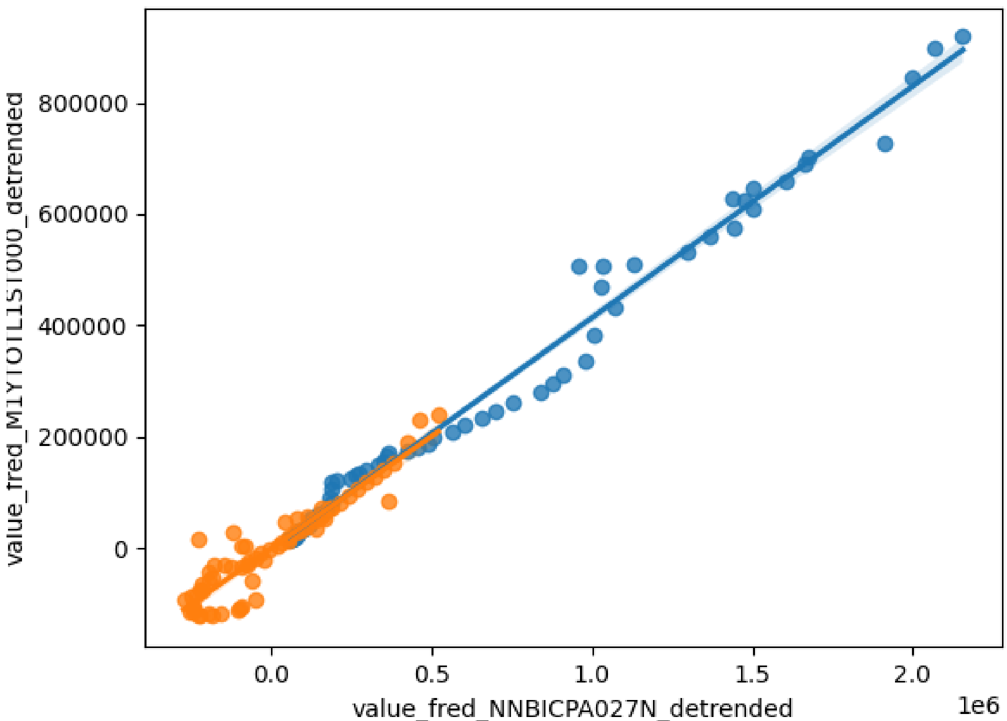


**Figure 4:** Detrended Regression Plot for 2026-07-22

# 4 Date: 2026-07-23

**Series ID: NENUCK9POP**

This series is titled Resident Population in Nuckolls County, NE and has a frequency of Annual. The units are Thousands of Persons and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1970-01-01 and the observation end date is 2025-01-01.The popularity of this series is 1.

**Series ID: COGANOPIBLEND**

This series is titled Medical Services Expenditures by Disease: Other: Congenital Anomalies Price Index, Blended Account Basis and has a frequency of Annual. The units are Index 2017=100 and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 2000-01-01 and the observation end date is 2021-01-01.The popularity of this series is 1.

## 4.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

| | | | |
|---|---|---|---|
| **Dep. Variable:** | value_fred_COGANOPIBLEND | **R-squared:** | 0.876 |
| **Model:** | OLS | **Adj. R-squared:** | 0.870 |
| **Method:** | Least Squares | **F-statistic:** | 141.6 |
| **Date:** | Thu, 23 Jul 2026 | **Prob (F-statistic):** | 1.58e-10 |
| **Time:** | 13:20:04 | **Log-Likelihood:** | -75.559 |
| **No. Observations:** | 22 | **AIC:** | 155.1 |
| **Df Residuals:** | 20 | **BIC:** | 157.3 |
| **Df Model:** | 1 | | |
| **Covariance Type:** | nonrobust | | |

| | **coef** | **std err** | **t** | **P> \|t\|** | **[0.025** | **0.975]** |
|---|---|---|---|---|---|---|
| **const** | 422.9442 | 28.506 | 14.837 | 0.000 | 363.483 | 482.406 |
| **value_fred_NENUCK9POP** | -75.7107 | 6.363 | -11.898 | 0.000 | -88.985 | -62.437 |

| | | | |
|---|---|---|---|
| **Omnibus:** | 4.239 | **Durbin-Watson:** | 1.347 |
| **Prob(Omnibus):** | 0.120 | **Jarque-Bera (JB):** | 2.271 |
| **Skew:** | 0.630 | **Prob(JB):** | 0.321 |
| **Kurtosis:** | 3.944 | **Cond. No.** | 79.9 |

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

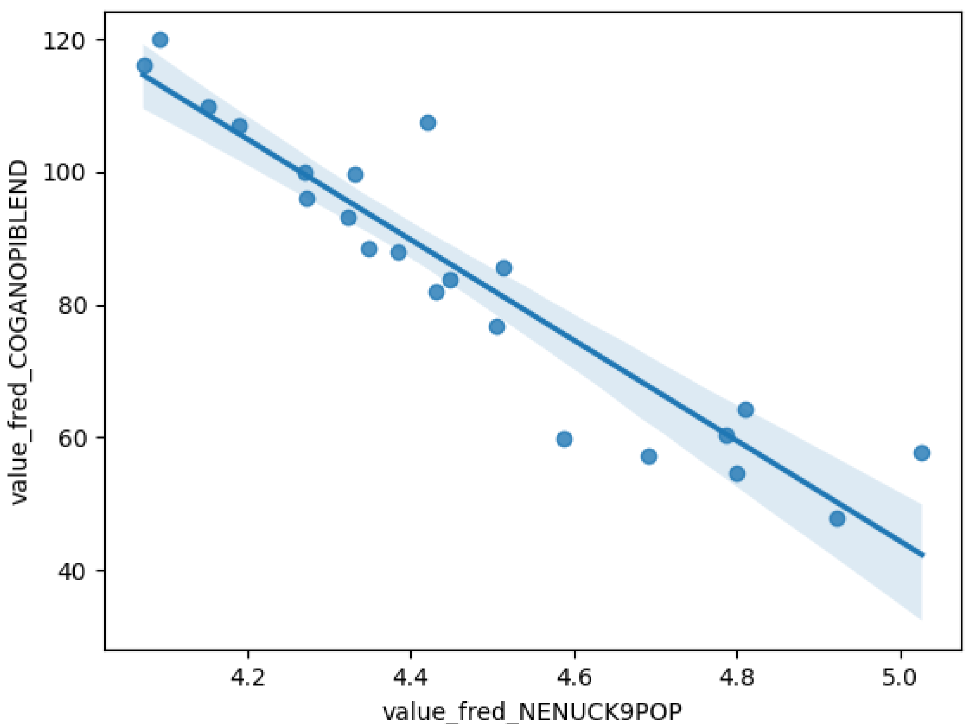


**Figure 5:** Raw Regression Plot for 2026-07-23

### 4.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

| | | | |
|---|---|---|---|
| **Dep. Variable:** | value_fred_COGANOPIBLEND_detrended | **R-squared:** | 0.358 |
| **Model:** | OLS | **Adj. R-squared:** | 0.326 |
| **Method:** | Least Squares | **F-statistic:** | 11.17 |
| **Date:** | Thu, 23 Jul 2026 | **Prob (F-statistic):** | 0.00324 |
| **Time:** | 13:20:05 | **Log-Likelihood:** | -75.552 |
| **No. Observations:** | 22 | **AIC:** | 155.1 |
| **Df Residuals:** | 20 | **BIC:** | 157.3 |
| **Df Model:** | 1 | | |
| **Covariance Type:** | nonrobust | | |

| | **coef** | **std err** | **t** | **P> \|t\|** | **[0.025** | **0.975]** |
|---|---|---|---|---|---|---|
| **const** | 6.695e-12 | 1.678 | 3.99e-12 | 1.000 | -3.499 | 3.499 |
| **value_fred_NENUCK9POP_detrended** | -78.2472 | 23.407 | -3.343 | 0.003 | -127.074 | -29.421 |

| | | | |
|---|---|---|---|
| **Omnibus:** | 3.865 | **Durbin-Watson:** | 1.355 |
| **Prob(Omnibus):** | 0.145 | **Jarque-Bera (JB):** | 1.985 |
| **Skew:** | 0.584 | **Prob(JB):** | 0.371 |
| **Kurtosis:** | 3.896 | **Cond. No.** | 14.0 |

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

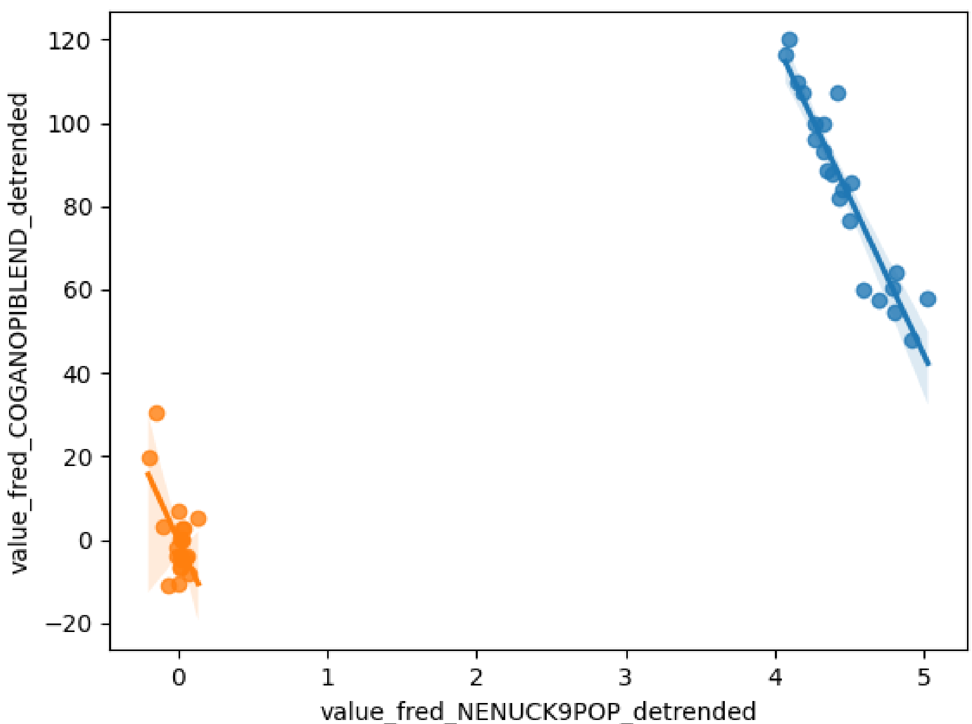


**Figure 6:** Detrended Regression Plot for 2026-07-23